MANAS RANJAN BARIK

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SUMMARY

Full Stack Developer with 1.8 years of experience building multi-tenant SaaS and AI-enabled applications using TypeScript, Node.js, React, NestJS, and PostgreSQL. Hands-on experience developing chat backend workflows, LLM integrations, tool execution, REST APIs, and access control for multi-tenant applications.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL
Frontend: React
Backend: Node.js, Express, NestJS, REST APIs, Microservices, LLM Integration, Tool Execution
Databases: PostgreSQL, Redis, Snowflake, DuckDB
Tools: Git, Prisma, TypeORM

PROFESSIONAL EXPERIENCE

DataCaffe.ai | Bangalore, India

Software Engineer **Dec 2024 – Present**

- Build backend services for Breww.ai, a multi-tenant AI chat application, using TypeScript, Node.js, NestJS, PostgreSQL, and REST APIs, supporting conversation processing, message persistence, and LLM integrations.
- Implement AI tool-execution workflows, enabling the chat backend to invoke application tools, process tool outputs, and return results through the conversational flow.
- Develop a document preview interface for Breww.ai using React and TypeScript, integrating document viewing directly into the chat experience.
- Develop per-user API usage and LLM token tracking with quota management, improving usage visibility and controlling resource consumption across customer accounts.
- Implement JWT authentication, Role-Based Access Control (RBAC), row-level security, and workspace isolation for 100+ users across multiple workspaces.
- Develop Snowflake and Excel data integrations with schema synchronization and automated retry logic, reducing failed data imports by 40%.

PROJECTS

RAG Tutor & Plagiarism Detector | Independent Project

Python, FastAPI, Embeddings, FAISS, Ollama, Streamlit

- Built a local AI dashboard combining a RAG-based science tutor and plagiarism analysis system using Python, FastAPI, Streamlit, FAISS, and Ollama.
- Developed a plagiarism pipeline using TF-IDF similarity, embedding-based paraphrase detection, and LLM-based AI-text classification to generate a 0–100 integrity score.
- Ran Q&A and analysis locally with Ollama models, grounding tutor responses in indexed textbook content and providing fallback handling when model services were unavailable.

ResBuild – ATS Resume Builder | Independent Project

Python, Flask, OpenAI API

- Built a Flask web application that parses PDF, DOCX, and TXT resumes, compares them against job descriptions using keyword-based ATS scoring, and generates single-column Word resumes tailored to target roles.
- Implemented offline and OpenAI-powered generation modes with before/after match scores, matched and missing keyword analysis, DOCX generation, and automatic fallback to local processing when the OpenAI service is unavailable.

Excel-to-Parquet Query Pipeline | Personal Project

DuckDB, Parquet, Excel

- Built a data-processing prototype that converted Excel sheets into Parquet files for more efficient local storage and querying.
- Queried the generated Parquet datasets using DuckDB to validate data access and transformation workflows.

EDUCATION

GITA Autonomous College, Bhubaneswar, Odisha
B.Tech, Electrical Engineering - **CGPA:** 7.71 **2019 – 2023**